

OPTIONS RESEARCH DISCIPLINE - ONE PAGE SUMMARY

Based only on repository evidence

WHAT IT IS

A personal, rule-based options research system with a Next.js interface and shared TypeScript engines. The UI frames outputs as research-only approximations and not trade recommendations.

WHO IT IS FOR

Primary persona: a self-directed options trader focused on disciplined paper trading and behavior review.

WHAT IT DOES

- Dashboard links Rules, Strategy Lab, Backtests, Gatekeeper, Paper Portfolio, Journal, and Behavior Insights.
- Defines risk limits: max risk/trade, max positions, daily loss, cooldown, ticker and strategy concentration.
- Evaluates candidate trades with ALLOWED/WARNINGS/BLOCKED status and compliance scores.
- Builds strategy templates using trend and volatility filters (SMA, RSI, breakout, realized volatility).
- Runs approximate backtests with equity curve, drawdown, win rate, loss streaks, and violation pressure.
- Tracks simulated positions and exposure analytics by ticker, strategy, and expiration clustering.
- Supports credentials auth plus Prisma models for users, rules, strategies, runs, trades, positions, and logs.

HOW IT WORKS (REPO-EVIDENCED ARCHITECTURE)

- Web UI: Next.js app in apps/web (App Router pages under src/app).
- Auth: NextAuth credentials provider with scrypt password verification.
- Data: Prisma client in web app and PostgreSQL schema in prisma/schema.prisma.
- Domain logic: packages/core (rule engine, compliance scoring, portfolio exposure helpers).
- Research logic: packages/research (Yahoo OHLCV fetch, indicators, candidate generation, backtest runner).
- Data flow: market bars -> strategy candidates -> rule checks -> simulated fills/backtest metrics -> persisted entities/logs.
- API route handlers wiring pages to packages: Not found in repo.

HOW TO RUN (MINIMAL GETTING STARTED)

- Install dependencies: npm install
- Configure database: set DATABASE_URL (required by Prisma datasource).
- Generate Prisma client: npx prisma generate
- Start the web app: npm run dev, then open <http://localhost:3000>
- Migration/seed instructions and production deployment guide: Not found in repo.